The resistance of the electrolytic solution (R) is calculated from the known resistance values P, Q and the measured 'S' value under balanced condition using the above expression (9.10).

Conductivity calculation

Specific conductance (or) conductivity of an electrolyte can be calculated from the resistance value using the following expression.

$$\kappa = \frac{1}{R}\left(\frac{l}{A}\right) \qquad [\because \text{ equation 9.5}]$$

The value of the cell constant $\frac{l}{A}$ is usually provided by the cell manufacturer. Alternatively the cell constant may be determined using KCl solution whose concentration and specific conductance are known.

Example

The resistance of a conductivity cell is measured as 190 Ω using 0.1M KCl solution (specific conductance of 0.1M KCl is 1.3 Sm^{-1}). When the same cell is filled with 0.003M sodium chloride solution, the measured resistance is 6.3KΩ. Both these measurements are made at a particular temperature. Calculate the specific and molar conductance of NaCl solution.

Given that

$\kappa = 1.3\ Sm^{-1}$ (for 0.1M KCl solution)

$R = 190\ \Omega$

$$\left(\frac{l}{A}\right) = \kappa \,.\, R = (1.3\ Sm^{-1})\ (190\ \Omega)$$

$$= 247\ m^{-1}$$

$$\kappa_{(NaCl)} = \frac{1}{R_{(NaCl)}}\left(\frac{l}{A}\right)$$

$$= \frac{1}{6.3\ K\Omega}\ (247\ m^{-1}) \qquad 6.3K\Omega = 6.3 \times 10^{3}\Omega$$

$$= 39.2 \times 10^{-3}\ Sm^{-1}$$

$$\Lambda_m = \frac{\kappa \times 10^{-3}\ mol^{-1}\ m^3}{M}$$

$$= \frac{39.2 \times 10^{-3}\ (Sm^{-1}) \times 10^{-3}\ (mol^{-1}m^3)}{0.003}$$

$$\Lambda_m = 13.04 \times 10^{-3}\ Sm^2\ mol^{-1}$$

9.2 Variation of molar conductivity with concentration

Friedrich Kohlraush studied the molar conductance of different electrolytes at different concentrations. He observed that, increase of the molar conductance of an electrolytic solution with the increase in the dilution. One such experimental results is given in the following table for better understanding.

XII U9 Electro Chemistry.indd 39 16-12-2022 19:36:55